

Should an AI Agent Call a Tool, or Refuse?

3,491

Agent Tasks

33

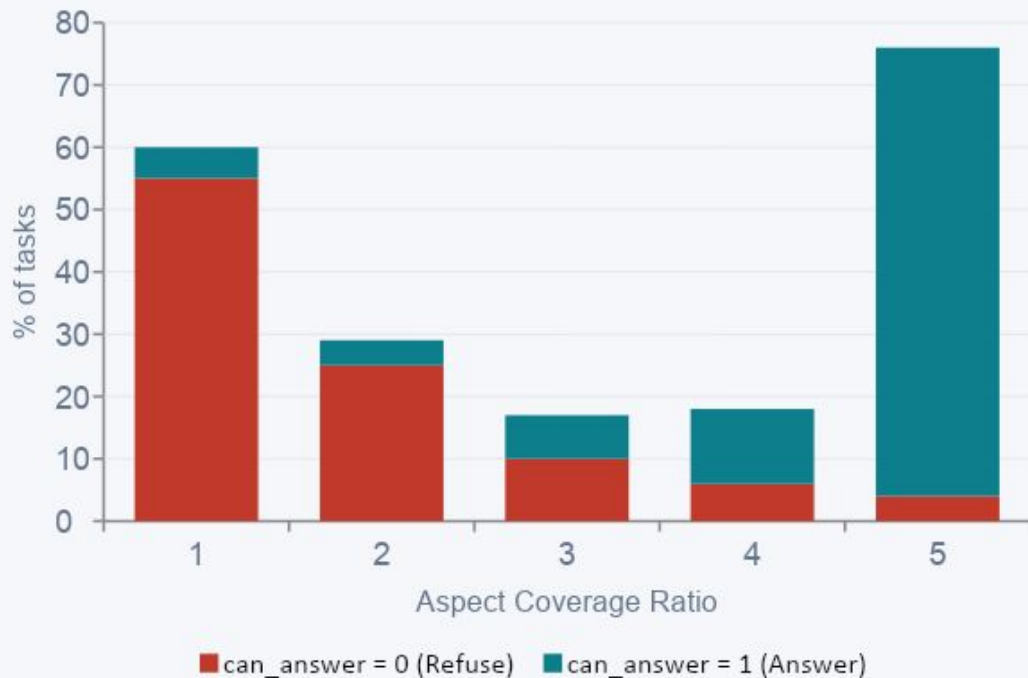
Features

2

Classes

can_answer: 0 | 1

Aspect Coverage Ratio Separates the Classes



Tasks that can be answered overwhelmingly have ratio near 1.0 — tools match the query topics.

Refusal tasks cluster at 0 — available tools are irrelevant to the query.

This single feature achieves strong class separation and becomes the most important model feature.

Random Forest — Test Set Results

95%

Accuracy

96%

Precision

98%

Recall

97%

F1-Score

Hyperparameters: `n_estimators = 200` · `max_depth = 20`

Model Comparison — Validation F1



False Positives Are the Costliest Mistake

False Positive

Predict CAN answer · Actually CANNOT

- Agent calls a tool that cannot help
- May perform harmful actions (e.g. send, book, delete)
- Wastes resources and erodes user trust
- Harder to recover from in production

Most Dangerous

False Negative

Predict CANNOT answer · Actually CAN

- Agent refuses a solvable request
- User experience suffers — unhelpful AI
- No harmful side-effect
- User can rephrase and retry

Recoverable

Preference: A conservative model (high precision) is preferred in production to prevent unsafe tool calls.

Tasks Cluster Naturally by Tool Relevance

K-Means (k=2)

Agglomerative (Ward)

Cluster 0

Low-Match / Refusal

Answer Rate ~5%

Aspect Coverage ~0.05

Top Domain mixed

Error Profile Mostly TN

Complexity low-medium

Cluster 1

High-Match / Answerable

Answer Rate ~98%

Aspect Coverage ~0.92

Top Domain travel / other

Error Profile Mostly TP

Complexity low

Dataset Limitation · Suggested Improvement

Dataset Limitation

The `can_answer` label is synthetically derived from benchmark construction rules — not from new human annotations. Models may learn patterns specific to the benchmark artifacts rather than truly generalising to real-world tool-use reasoning.

Suggested Improvement

Replace strict keyword-based aspect matching with embedding-based cosine similarity between the query and tool descriptions. This would capture semantic equivalences (synonyms, paraphrases) that keyword rules miss, creating far richer and more robust features.